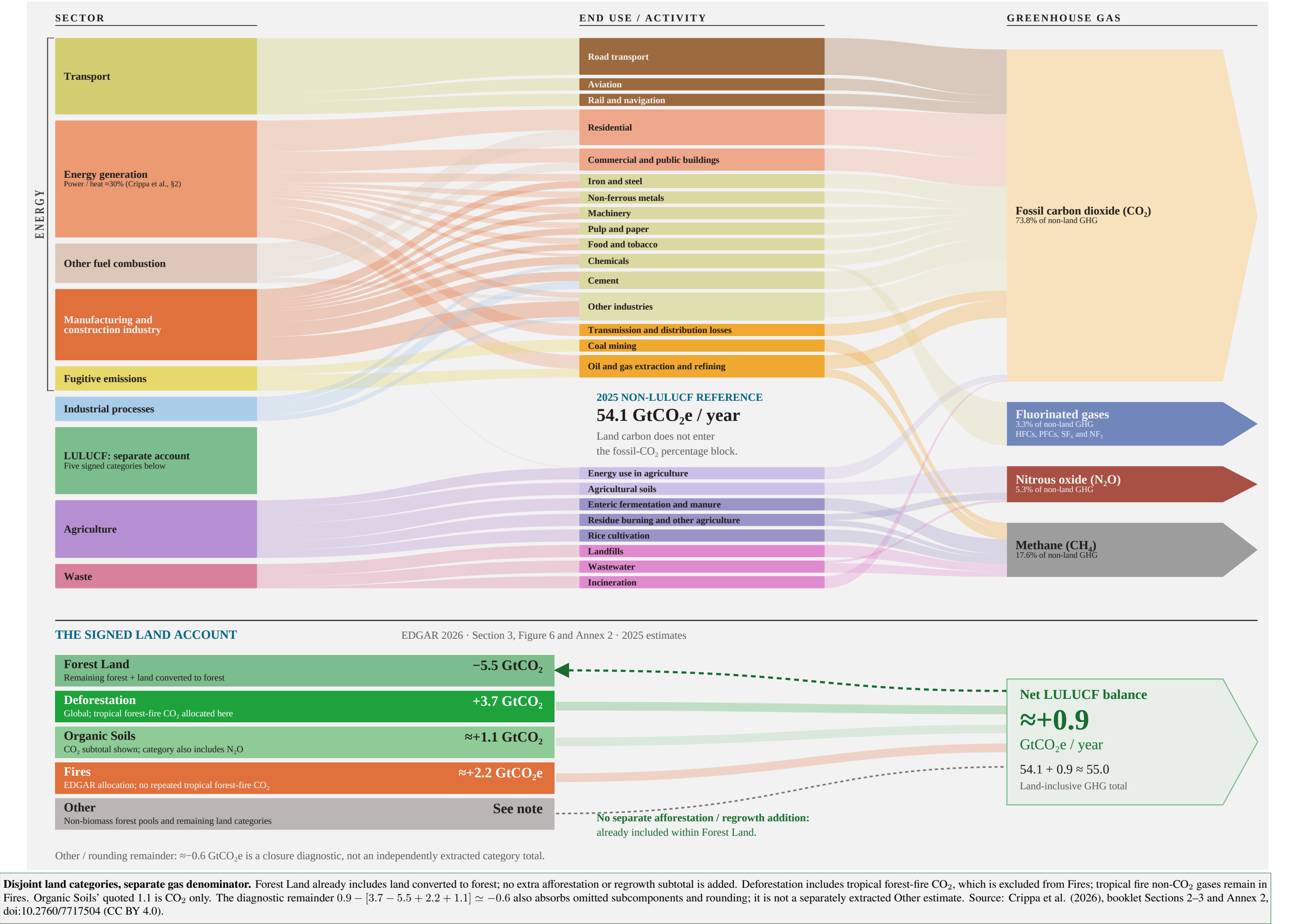
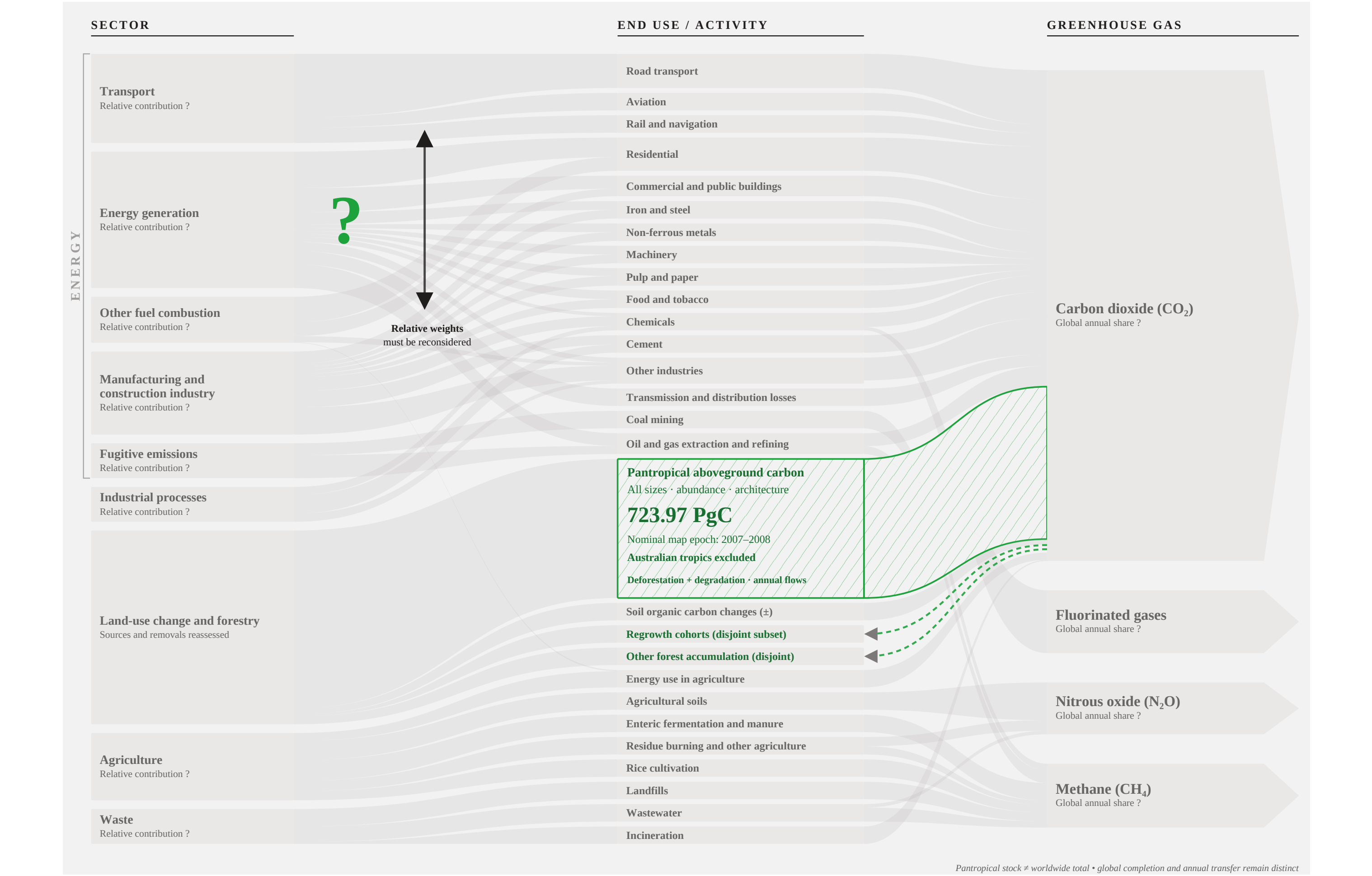






All sizes, not only the largest trees. In the *Acalypha–Guazuma* community, the 10–30 cm class contains 38.97% of reported biomass in a community with total 147.8 Mg ha^{−1}; 2.88 t divided by 30 stems gives a derived mean of **96 kg biomass per stem** (Vásquez and Arellano, 2012, Table 170, p. 942). This older class distribution is allometry-based and illustrates abundance rather than class-5 density; the separate architectural records near 100 kg illustrate that method’s size coverage. Abundance, affected fractions and release histories govern population contribution.

Physical consequence and attribution. A positive net source missing from an account leaves actual accumulation above the account’s reconstruction unless other terms offset it. With total and removals held fixed, forest loss receives a larger share and can become the leading source if its corrected flux exceeds competing categories. Delayed release enters later years. Observed air can already contain emissions missed by an inventory; changing attribution does not itself establish an equally large measurement error in atmospheric concentration.

SPATIAL COVERAGE: THE PANTROPICAL RESULT IS NOT A WORLDWIDE TOTAL

Mapped contribution: 723.97 PgC
Aboveground carbon; Australian tropics excluded.

Nominal output-map reference: **2007–2008** (preprint p. 23). Baccini’s **2012** publication describes a 2007–2008 map; Simard’s height predictor uses 2005 GLAS data. Forest-only totals require a common forest mask.

HISTORICAL 2010 ILLUSTRATION — NOT A 2025 EMISSIONS ESTIMATE

Fixed total: 26.16–32.61%. Other contributions retained: 22.58–26.66%. Both impose the stock multiplier on Ecofys’s combined land-use category. The published 26.15–32.60% endpoints are reproduced by truncation; alternative hidden precision is possible. Supplementary Note S3 gives the source, assumptions and calculation.

Additional regional stock
Australian tropics and extratropical vegetation require separate estimates.

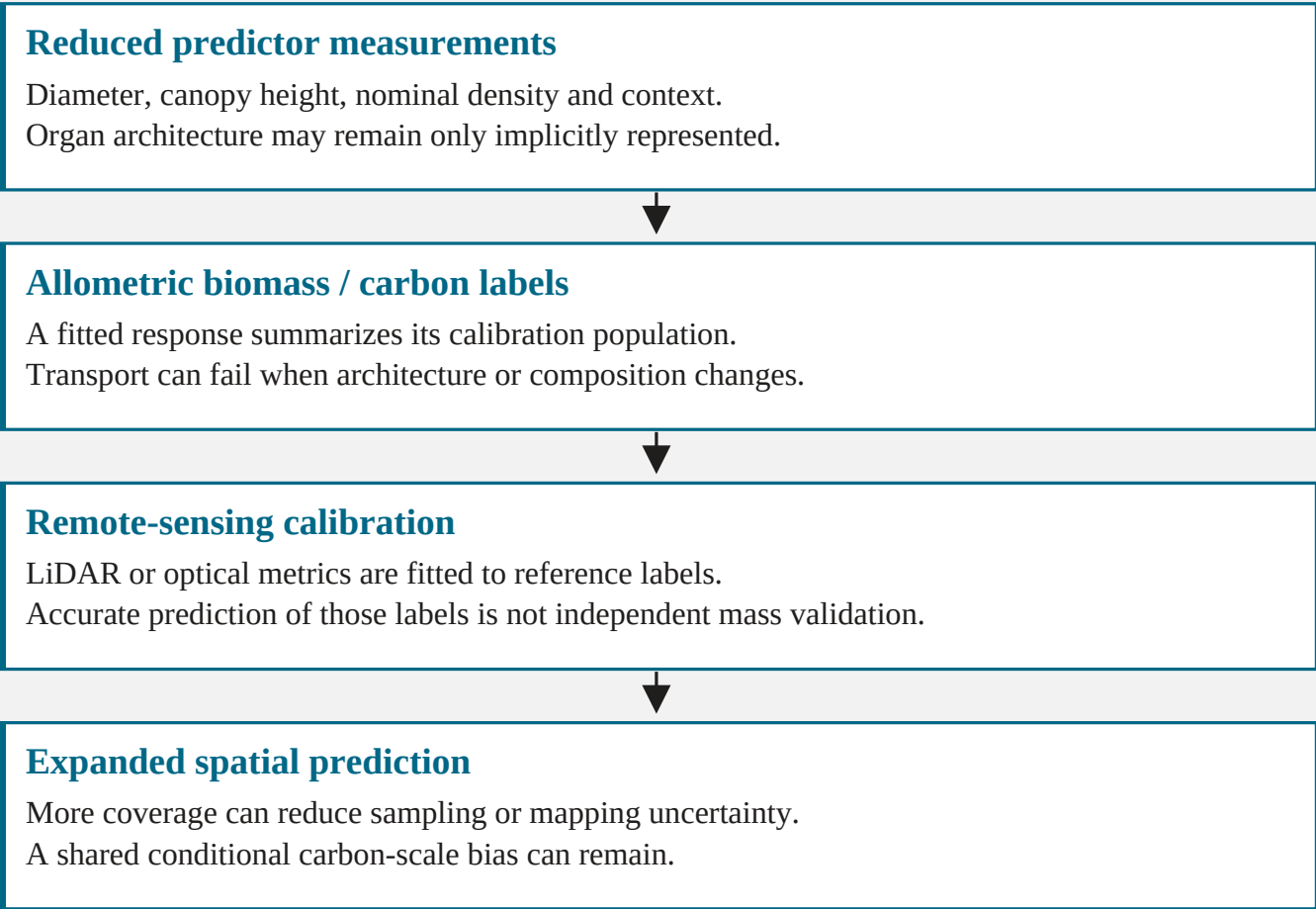
Worldwide completion
Add compatible pools and dates; annual net exchange remains separate.

Figure 2: Pantropical architectural measurement and worldwide annual attribution. The green branch identifies 723.97 PgC as the mapped pantropical aboveground-carbon estimate, explicitly excluding the Australian tropics (Arellano-P. and Rangel-Ch., 2016, pp. 23–24). Its 2007–2008 nominal map epoch differs from publication and predictor-acquisition dates (Baccini et al., 2012; Simard et al., 2011). Global coverage additionally requires omitted tropical and extratropical vegetation on matched pool, mask and date definitions. Hatching denotes unresolved annual transfer; the label retains the published standing stock, and graphical areas are schematic. The organization adapts the WRI/Ecofys source–activity–gas convention. Regrowth and other accumulation are disjoint conceptual subsets, not extra EDGAR subtotals. Muted sectors, the question mark and opposing arrows express a larger source total and reconstructed atmospheric burden under fixed other terms, a larger forest-loss attribution within a retained account, or joint reassessment. All stem sizes and partial degradation are included; no source ranking is inferred from graphical width. The historical inset is conditional 2010 arithmetic, not a measured 2025 allocation. Figure 3 separates allometry-calibrated spatial prediction from architecture-informed material measurement. The article distinguishes measurement evidence from annual-transfer conditions; source-table and class-level comparison details are in the technical supplement.

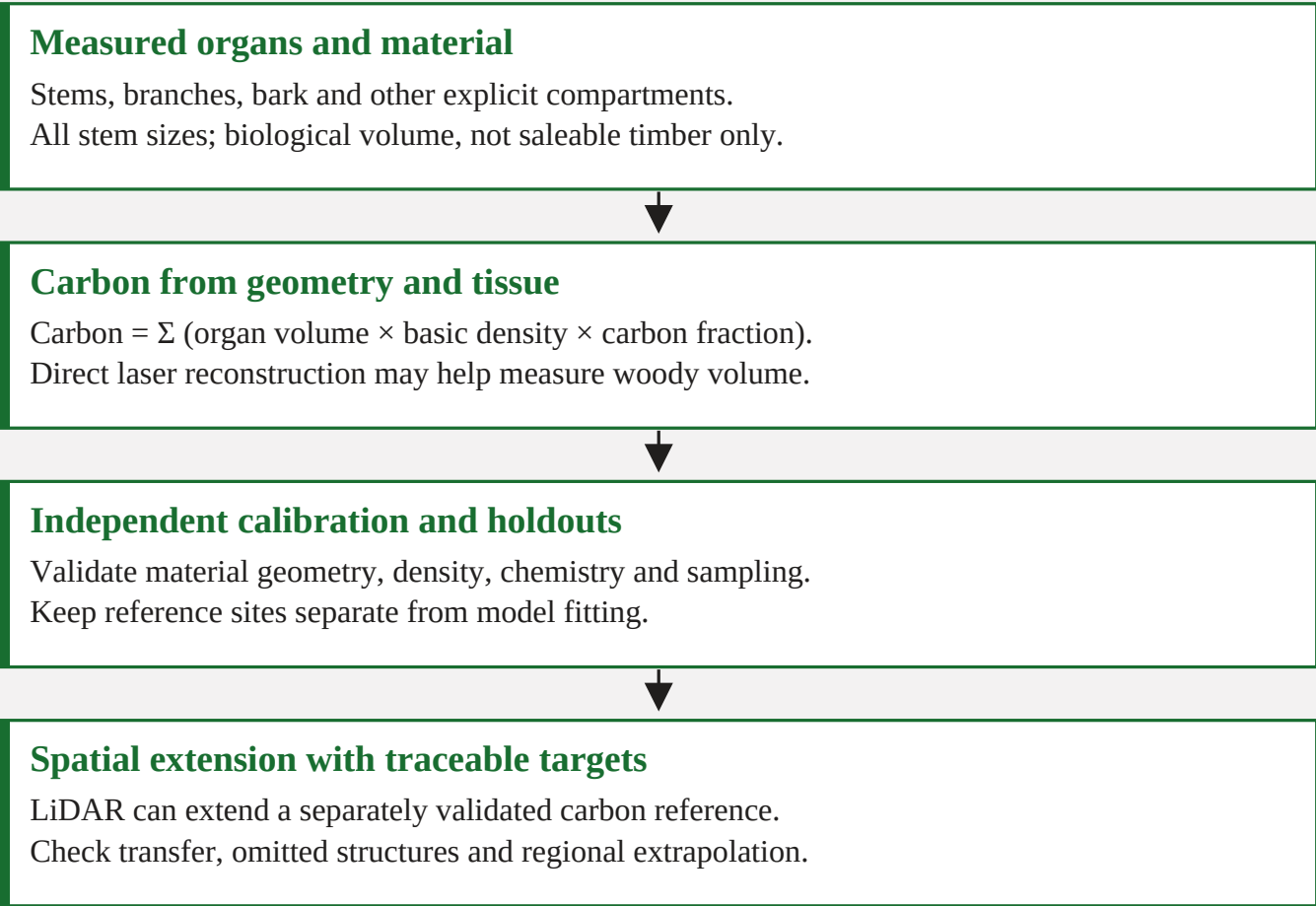
CARBON ACCURACY STARTS WITH THE REFERENCE

Spatial coverage measures where; a validated material conversion determines how much.

A • PROXY-LABEL PATHWAY



B • ARCHITECTURE-INFORMED PATHWAY



INHERITANCE IDENTITY • IDEAL CONDITIONAL PREDICTION

$$E[\text{label} \mid \text{LiDAR}] - E[\text{true carbon} \mid \text{LiDAR}] = E[\text{reference-label bias} \mid \text{LiDAR}]$$

Conditional mean-zero residual noise is assumed. This is a mechanism, not an empirical global correction factor.

Published methodological application

723.97 PgC · mapped pantropical aboveground carbon
Nominal map epoch 2007–2008; Australian tropics excluded.
221 reconstructed trees; 2,232 tissue samples reported.
Physical-volume checks and spatial calibration are distinct steps.

From carbon measurement to annual accounts

All sizes × abundance × affected fractions × release histories.
Deforestation, degradation, regrowth and other removals.
Uncounted net release raises reconstructed accumulation.
A retained total constrains reattribution among sources.

NOT EVERY LiDAR PRODUCT USES THE SAME MEASUREMENT CHAIN

Proxy-based biomass mapping and direct terrestrial 3D material reconstruction are different pathways.

Measure architecture and organ allocation to identify the source of allometric calibration error.

Refining calibration can improve the stock estimate without changing the distinction between reference accuracy and coverage.

Sources: Arellano-P. & Rangel-Ch. (2016); Goodman et al. (2014); Calders et al. (2015); GEDI L4A V2 documentation.
Operational diagram: no graphical width represents a measured carbon share. Henry Arellano-Peña.

The calibration distinction. A remote-sensing product fitted to allometric reference labels can inherit their conditional bias. Direct reconstruction of biological material with measured tissue properties provides a different reference pathway, which can itself use terrestrial laser measurements. Independent reference tests are required for both approaches; greater coverage alone does not determine absolute carbon accuracy.

Figure 3: Two measurement-to-map pathways and their annual-accounting consequences. The left pathway extends a calibrated proxy; the right begins with organ geometry and matched material properties. Architecture and tissue variation identify mechanisms by which an allometric predictor can depart from actual biological carbon. The conditional-expectation identity assumes a source distribution and mean-zero residual noise conditional on the sensor variables; a real learner also has fitting and transfer errors. The original reconstruction and physical tests are documented by Arellano-P. and Rangel-Ch. (2016). Crown-related biomass discrepancies are independently illustrated by Goodman et al. (2014). Calders et al. (2015) demonstrates a terrestrial laser material-volume route, while the GEDI L4A Version 2 guide documents a model linking waveform height metrics and biomass reference data. These examples retain their study domains and establish no universal multiplier. Total material volume includes nonmerchantable woody organs; belowground and dead pools remain separately defined. The reported 723.97 PgC is a geographically and temporally delimited standing stock. Its annual implications require affected populations, release histories and removals. Neither larger spatial coverage nor numerical reproduction of calibration labels is equivalent to independent carbon-mass validation. Original diagram: Henry Arellano-Peña.